

# PipeFit Pro — User Guide

Everything in this app does one job: turn what you can **measure** into what you have to **cut**.

This guide goes module by module. For each one: what it is for, what to measure, what to type, what comes back, **why it comes back that way**, and how to check it against your own head before you cut steel.

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# Part 1 — Read this first

## The one thing the whole app is about

A drawing gives you **centre-to-centre**. A saw needs **cut length**. They are never the same number, because a fitting occupies some of the distance between two centres.



**Cut = centre-to-centre – takeout A – takeout B – any weld gaps.**

Every module in this app is that sentence, worked for a different shape of job. Once you see it, the app stops being ten calculators and becomes one.

## Where the numbers come from

Nothing here is a rule of thumb someone remembered.

| KIND OF NUMBER     | SOURCE                                                           |
|--------------------|------------------------------------------------------------------|
| Fitting takeouts   | The Lindsey Pipefitters Handbook tables, transcribed and checked |
| Pipe OD and wall   | ASME B36.10M                                                     |
| Butt weld fittings | ASME B16.9                                                       |
| Flanges            | ASME B16.5                                                       |
| Screwed fittings   | ASME B16.3                                                       |
| Pipe threads       | ASME B1.20.1                                                     |
| Miter code limit   | ASME B31.3 §304.2.3                                              |
| Bend geometry      | The bend radius, worked exactly                                  |

**1,370 automated tests** run on every change. Ten misprints were found in the printed handbook while transcribing it, and each one is pinned by a test so it cannot quietly come back.

## Why you can trust it without second-guessing

The app is checked against the multipliers a fitter already carries in his head. Multiply the offset by these:

| FITTING ANGLE | TRAVEL         | RUN            | SHRINK         |
|---------------|----------------|----------------|----------------|
| 11.25°        | × 5.126        | × 5.027        | × 0.098        |
| 22.5°         | × 2.613        | × 2.414        | × 0.199        |
| 30°           | × 2.000        | × 1.732        | × 0.268        |
| <b>45°</b>    | <b>× 1.414</b> | <b>× 1.000</b> | <b>× 0.414</b> |
| 60°           | × 1.155        | × 0.577        | × 0.577        |
| 90°           | × 1.000        | × 0            | × 1.000        |

The app agrees with every one of them to **four decimal places**, and a test proves it on every build. If it ever disagrees with that table, the app is wrong — not the table.

## Set three things once

**Settings** (gear, top right of any screen).

| SETTING               | SET IT TO                                            | WHY                                                                                                           |
|-----------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Default size</b>   | The pipe you are on today                            | Every screen opens on it. Saves a tap on every job.                                                           |
| <b>Weld gap</b>       | Your standard root gap, default 3/32"                | Comes off welded and flanged ends only. A screwed or soldered joint pulls up tight and never has it deducted. |
| <b>Fractions</b>      | 1/16 for most work, 1/32 or 1/64 for tight tolerance | The decimal above the fraction is always the exact figure. Cut to the decimal when it matters.                |
| <b>Length readout</b> | Inches                                               | The calculator stays in inches. Press <b>FT</b> when you actually want feet.                                  |

# Part 2 — The three ideas everything is built on

## 1. Takeout — why a fitting "eats" length

An elbow does not start at the centre point. The centre point is where two lines cross in the air; the fitting's face is some distance back from it. That distance is the **takeout**.

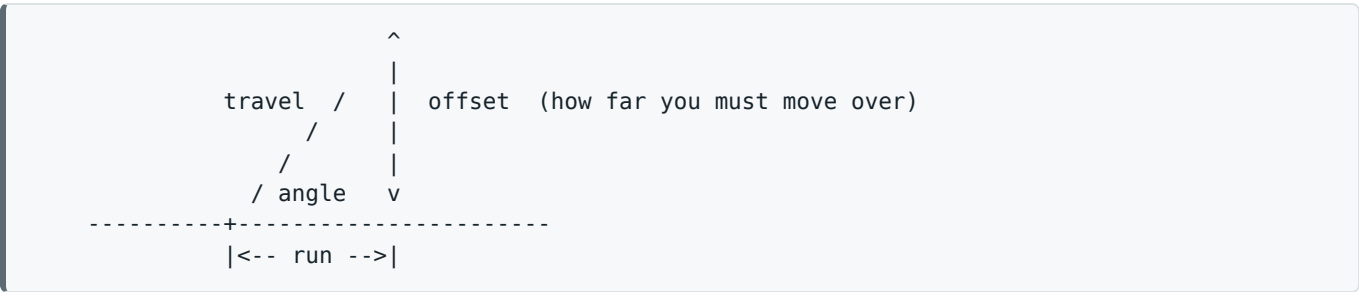
For a **bend you make yourself**, the takeout is pure geometry:

takeout = bend radius × tan(half the angle)

A 2" long radius elbow bends on a 3" radius (1.5 × the size). At 90°, tan 45° = 1, so the takeout is **3"**. At 45°, tan 22.5° = 0.4142, so it is **1.2426"**.

For a **fitting you buy**, the takeout is whatever the manufacturer made it, out of the handbook table.

## 2. Every offset is a right triangle



- **Offset** — how far the pipe has to shift. You measure this.
- **Run** — how far it advances along the original line while it shifts.
- **Travel** — centre to centre of the two fittings. **This is the number the cut comes from.**
- **Shrink** — travel minus run. What the offset costs you in length.

Pick any two and the third follows. That is all the offset screens do.

## 3. A bend is NOT a bought fitting

They agree at 90° and disagree everywhere else.

| 2" LONG RADIUS, 45° | TAKES OUT |
|---------------------|-----------|
| Pipe you bent       | 1.2426"   |
| Elbow you bought    | 1-3/8"    |

A run of bought fittings worked to the bend formula cuts **every piece an eighth of an inch long**.

**The app keeps them separate on purpose. Cut length** offers both, labelled. **Simple offset, Rolling offset** and **Pipe bend** work the bend, because that is what a bender does.

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## Part 3 — The modules

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### Simple offset

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**Get around an obstruction in one plane.** The pipe moves one way — up, down, left or right — and carries on.

#### What to measure

1. **Offset** — how far the pipe must shift to clear the obstruction, centre to centre.
2. Either the **fitting angle** you intend to use, or the **run** you have room for. Not both — the app solves whichever you do not give it.

#### Step by step

1. Tap **Simple offset**.
2. Set the pipe with the wrench button at the bottom (size, LR/SR, schedule).
3. Type the **Offset**.
4. Tap an angle preset — **45°** is the one you will use most.
5. Read **PIPE CUT**.

#### What each number means

| RESULT                          | WHAT IT IS                           | WHY YOU CARE                                |
|---------------------------------|--------------------------------------|---------------------------------------------|
| <b>PIPE CUT</b>                 | The piece between the two fittings   | This is what you cut                        |
| <b>Travel</b>                   | Centre to centre of the two fittings | The cut before deductions                   |
| <b>Run</b>                      | How far it advances along the line   | Tells you if it fits the space you have     |
| <b>Shrink</b>                   | Travel – run                         | What the offset costs you in overall length |
| <b>Setback</b>                  | Takeout per fitting                  | What comes off each end                     |
| <b>Cut angle</b>                | The angle the fittings turn through  | Confirms the preset took                    |
| <b>Inside arc / Outside arc</b> | Throat and back of the bend          | Marks for bending, not for buying           |

#### Worked example

**2" schedule 40, long radius, 12" offset at 45°, default 3/32" gap:**

| FIGURE             | VALUE           |
|--------------------|-----------------|
| Run                | 12.0000"        |
| Travel             | 16.9706"        |
| Shrink             | 4.9706"         |
| Setback (each end) | 1.2426"         |
| <b>PIPE CUT</b>    | <b>14.2978"</b> |

**Check it in your head:**  $12 \times 1.414 = 16.97$  travel ✓. Two takeouts of  $1.24 = 2.49$ .  $16.97 - 2.49 = 14.48$ , less two  $3/32$ " gaps (0.19) = **14.29** ✓

### The square jog — 90°

Preset **90°**. The pipe goes straight across and advances nothing. **Run = 0, travel = the offset.**

**24" offset, 2" pipe:** run 0, travel 24.00, setback 3.00 each end → **PIPE CUT 17.81"**

### When the run is fixed instead

Tap **Set run** and type the run. The angle falls out of it and shows as **Cut angle**. A run of **0** gives you the square jog.

Use this when the space along the line is what is tight, not the angle.

### Where people go wrong

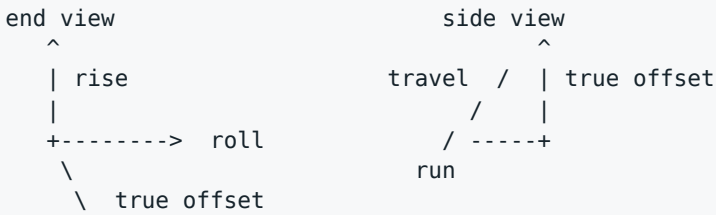
- **Measuring the offset to the pipe surface instead of the centre.** Everything in this app is centre to centre.
- **Forgetting the gap is already off.** PIPE CUT is ready to cut. Do not deduct again.
- **Using a 45° preset while installing bought 45° elbows.** This screen works a *bend*. For bought fittings use **Cut length**, which has both.

## Rolling offset

The obstruction makes the pipe move two ways at once — up *and* over. A rolling offset is a simple offset that has been rotated around the pipe.

### Why it is two triangles, not one

Look down the pipe from the end. The **rise** (up/down) and the **roll** (left/right) are two legs of a right triangle. Their hypotenuse is the **true offset** — the real, straight-through distance the pipe has to move.



Then the **true offset** and the **run** form a second right triangle whose hypotenuse is the **travel**. Same theorem twice. That is the whole of it.

## Step by step

1. Tap **Rolling offset**.
2. Set the pipe.
3. Type **Rise / set** (up or down) and **Roll** (left or right).
4. Tap an **Elbow angle** preset, or tap **Run** to fix the run instead.
5. Read **PIPE CUT**.

## What each number means

| RESULT             | WHAT IT IS                                                               |
|--------------------|--------------------------------------------------------------------------|
| <b>True offset</b> | The real diagonal distance, straight through                             |
| <b>Roll angle</b>  | How far around the pipe the offset lies — which way to point the fitting |
| <b>Travel C2C</b>  | Centre to centre of the two fittings                                     |
| <b>Run</b>         | Advance along the original line                                          |
| <b>PIPE CUT</b>    | The piece between them                                                   |

**The roll angle is the one people miss.** It is not a cut dimension — it tells you how far to rotate the fitting around the pipe so the offset lands where you measured it.

## Worked example

**2" schedule 40 LR, rise 12", roll 9", 45° elbows:**

| FIGURE          | VALUE           |
|-----------------|-----------------|
| True offset     | 15.0000"        |
| Roll angle      | 36.87°          |
| Run             | 15.0000"        |
| Travel C2C      | 21.2132"        |
| Setback         | 1.2426"         |
| <b>PIPE CUT</b> | <b>18.5404"</b> |

**Check it:** 12 and 9 is a 3-4-5 triangle scaled by three — true offset **15** ✓. At 45° the run equals the true offset, 15 ✓. Travel =  $15 \times 1.414 = 21.21$  ✓

## Where people go wrong

- **Entering the true offset in the rise box.** Rise and roll are the two *separate* measurements. The app works the true offset out.
  - **Ignoring the roll angle.** Get it wrong and the spool is geometrically perfect and points the wrong way.
  - **Rise and roll both zero** — there is no offset to solve, and the app says so.
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## Cut length

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**The most used screen in the app.** You have a centre-to-centre dimension off a drawing and a fitting on each end.

### Step by step

1. Tap **Cut length**.
2. Set the pipe.
3. Type **C2C length**.
4. Pick **Joint** — screwed, welded, flanged or soldered.
5. If flanged, pick the **Class** — 150, 300, 600 and up.
6. Pick **End A** and **End B**.
7. Read **PIPE CUT**.

### Why the joint family matters

It changes two things at once:

1. **Which table the takeout comes from.** A screwed 90 and a welded 90 in the same size are different castings with different dimensions.
2. **Whether the weld gap is deducted.** Welded and flanged ends leave a root gap. Screwed and soldered joints pull up tight and never have one deducted.

That second one was a real bug once. It is now a rule in the code and a test.

### Worked example

**24" C2C, 2" schedule 40, welded long radius 90 on both ends:**

| FIGURE          | VALUE           |
|-----------------|-----------------|
| End A takeout   | 3.0000"         |
| End B takeout   | 3.0000"         |
| Total deduction | 6.0000"         |
| <b>PIPE CUT</b> | <b>18.0000"</b> |
| Weight          | 5.48 lb         |

With a 1/8" gap on both ends: **17.75"**

### Anything not on the list

Pick **Custom** on that end and type the takeout yourself. Use it for a valve, a strainer, a specialty fitting, or anything a vendor drawing gives you directly.

### Where people go wrong

- **Leaving the joint on the wrong family.** Check the **Joint** chip before you read the number.
- **Assuming a 45 is a 45.** A bought 45 and a bent 45 have different takeouts. Both are on the list, labelled.
- **Not setting the class on flanged.** A 150 and a 300 flange are not the same length.

## Thread engagement

**How far a threaded joint pulls up**, and what that does to your cut.

### Why a threaded joint is different

On a welded joint the pipe stops at the fitting face. On a threaded joint the pipe **screws into** the fitting — so the pipe reaches past the face, and the deduction is smaller than the fitting's centre-to-face.

**Deduction per end = fitting centre-to-face – total engagement**

Engagement has two parts:

- **Hand tight (L1)** — how far it goes in by hand. From ASME B1.20.1.
- **Wrench makeup** — how far your wrench turns take it further. Each turn advances the pipe by one **pitch**, and pitch is  $1 \div \text{threads per inch}$ .

### Step by step

1. Tap **Thread engagement**.
2. Tap the **Size**.
3. Set **Wrench turns** if you pull up tighter or looser than standard.
4. Type **C2C length** to get the cut.

## Worked example

1" NPT, standard 3 wrench turns, 24" centre to centre:

| FIGURE                   | VALUE           | WHERE IT COMES FROM    |
|--------------------------|-----------------|------------------------|
| Threads per inch         | 11.5            | from the standard      |
| Pitch                    | 0.0870"         | $1 \div 11.5$          |
| Hand tight (L1)          | 0.4000"         | from the standard      |
| Wrench makeup            | 0.2609"         | $3 \times 0.0870$      |
| <b>Total engagement</b>  | <b>0.6609"</b>  | hand tight + wrench    |
| Thread remaining         | 0.3236"         | what is left unused    |
| Fitting centre-to-face   | 1.5000"         | B16.3 90° elbow        |
| <b>Deduction per end</b> | <b>0.8391"</b>  | $1.5000 - 0.6609$      |
| <b>PIPE CUT</b>          | <b>22.3217"</b> | $24 - 2 \times 0.8391$ |

**Watch "Thread remaining".** If it goes negative the joint is over-threaded — the pipe bottoms out before it seals.

## Where people go wrong

- **Using the welded takeout on screwed pipe.** It is a different, bigger number and the piece comes out short.
- **Not adjusting wrench turns to how you actually pull up.** Two extra turns on 1" pipe moves the cut by about 3/16" over two ends.

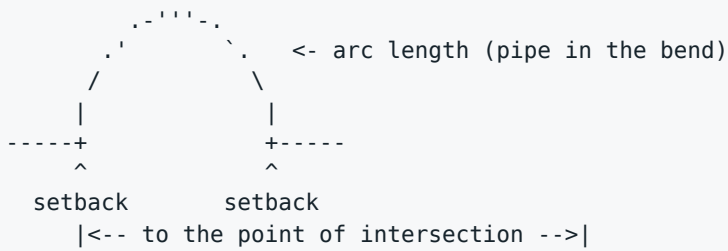
## Pipe bend

**Any angle, any radius, on your own bender.** This is where the marks come from.

### Why bending needs different numbers from cutting

When you bend, the pipe does not go round the corner — it cuts the corner on an arc. Three numbers describe that:

- **Setback** — from the point where the two legs would cross, back to where the bend starts.  $\text{radius} \times \tan(\text{half the angle})$ .
- **Arc length** — how much pipe is actually inside the bend.
- **Gain** — tangent total minus arc. **What you save by bending instead of mitring the corner.** Forget it and every piece comes out long.



## Step by step

1. Tap **Pipe bend**.
2. Pick the **Size**, then a **Radius** rule — 3D tight, 5D common field minimum, 10D long sweep — or type a radius.
3. Type the **Angle**.
4. Type **Leg A** and **Leg B**, measured **to the point of intersection**.
5. Set **Stock** if you want to know how many legs come out of a length.
6. Read the marks.

## Worked example

**2" pipe, 5D radius (10"), 45°, legs 30" and 30":**

| FIGURE                | VALUE           |
|-----------------------|-----------------|
| Setback (per tangent) | 4.1421"         |
| Arc length            | 7.8540"         |
| Gain                  | 0.4303"         |
| Tangent total         | 8.2843"         |
| <b>Mark from end</b>  | <b>25.8579"</b> |
| End of bend           | 33.7118"        |
| Piece length          | 59.5697"        |

**Check it:** setback =  $10 \times \tan 22.5^\circ = 4.142 \checkmark$ . Mark =  $30 - 4.142 = 25.858 \checkmark$ . Arc =  $10 \times 45^\circ$  in radians =  $10 \times 0.7854 = 7.854 \checkmark$

**At 90°, 5D radius, legs 24" and 18":** setback 10.00, arc 15.71, gain 4.29, mark from end 14.00, piece 37.71.

## Springback

If your bender overbends, put your known springback in and the app gives you the **overbend angle** to pull to. Leave it at zero if you have already dialled it out.

## Saddle bend

**Jump over a pipe or a beam without changing the line the conduit runs on.** It goes up, over, and back down to exactly where it was.

### Three point or four point

- **Three point** — for a round obstruction. One centre bend with two half-angle bends either side.
- **Four point** — for a wide obstruction. Two offsets back to back with a flat run over the top. Needs the **Width**.

### Why the centre bend is double the side bends

The conduit has to come back to the same line it started on. If the centre bend turns it through  $45^\circ$ , each side bend must turn it back through half that —  $22.5^\circ$  — for the two ends to end up parallel.

### Step by step

1. Tap **Saddle bend**.
2. Pick **Three point** or **Four point**.
3. Type the **Depth** — how far it has to rise to clear.
4. Type **To obstruction** — from the conduit end to the centre of the obstruction.
5. On a four point, type the **Width**.
6. Pick the centre bend angle.
7. Bend at the marks, in order.

### Worked example

**Three point, 2" deep, 20" from the end,  $45^\circ$  centre bend:**

| FIGURE               | VALUE        |
|----------------------|--------------|
| Side bend angle      | $22.5^\circ$ |
| Multiplier           | 2.6131       |
| Shrink per bend pair | 0.3978"      |
| Total shrink         | 0.7956"      |
| Developed length     | 10.4525"     |
| Minimum clearance    | 4.8284"      |

| MARK   | AT      | BEND  | NOTE                              |
|--------|---------|-------|-----------------------------------|
| Mark 1 | 15.172" | 22.5° | Bend up, arrow to the centre mark |
| Centre | 20.398" | 45°   | Bend over the obstruction         |
| Mark 3 | 25.624" | 22.5° | Bend down, back to level          |

**Check it:** spacing = depth × multiplier =  $2 \times 2.6131 = 5.226$ ". Centre mark is at  $20 + 0.398$  shrink = 20.398, and the outer marks sit 5.226 either side ✓

**Minimum clearance** is the one to watch. If the obstruction is closer to the end than that figure, the first bend would fall off the end of the conduit — and the app refuses rather than giving you a mark you cannot make.

## Miter bend

**Make an elbow out of straight pipe** when you do not have a fitting, or the size does not come as one.

### How the cut angle is worked out

With  $n$  segments you make  $n - 1$  cuts. Each cut face turns the pipe by **twice** the angle you cut it at, because both halves tilt. So:

$$\text{cut angle} = \text{total turn} \div (2 \times \text{number of cuts})$$

A 90° elbow in 4 segments is 3 cuts, so each cut is  $90 \div 6 = 15^\circ$ .

The two **end segments** have one mitred face. The **middle segments** have two, so their total included angle is double.

**Throat** is the short side of a segment, **back** is the long side. They come from the centreline radius minus and plus half the pipe OD.

### Step by step

1. Tap **Miter bend**.
2. Type the **Total turn** and pick the **Segments**.
3. Set the **Centreline radius**.
4. Mark and cut throat and back lengths on each segment.

### Worked example

**90° total, 4 segments, 2" schedule 40, 6" centreline radius:**

| FIGURE           | VALUE                     |
|------------------|---------------------------|
| Cuts             | 3                         |
| <b>Cut angle</b> | <b>15.00°</b>             |
| End segment      | 15.00° (one mitred face)  |
| Mid segment      | 30.00° (two mitred faces) |
| Throat length    | 2.5790"                   |
| Back length      | 3.8518"                   |
| Cutback          | 0.6364"                   |
| Centreline arc   | 9.4248"                   |

**Check it:** 2" pipe OD is 2.375", so half is 1.1875. Throat =  $2 \times (6 - 1.1875) \times \tan 15^\circ = 2 \times 4.8125 \times 0.2679 = 2.579$  ✓

### The code check

Above a **22.5° cut angle** the app warns you: ASME B31.3 §304.2.3 requires design verification for miters above that. It does not stop you — it tells you, because that is an engineering decision, not a field one.

More segments = smaller cut angle = a smoother, stronger elbow. If you get the warning, add a segment.

## 3D spool

**Build a whole run and look at it before you cut any of it.**

### How a spool is described

Each leg turns off the one before it by two numbers:

- **Bend** — how far it turns (the elbow angle).
- **Roll** — how far around the previous leg's axis that turn is pointed.

Roll is the part people find odd. With a 90° bend and 0° roll the pipe turns up. Same 90° bend at 90° roll and it turns sideways instead. Bend says *how much*, roll says *which way*.

### Step by step

1. Tap **3D spool**.
2. Set the pipe and the gap.
3. Set the first leg's **Length**.
4. **Add leg**, then give it a length, a **Bend** and a **Roll**.
5. Drag the picture to turn it. **NE / NW / SW / SE** jump to the four isometric corners.
6. **Hold a leg in the picture and drag along it** to stretch or shorten it.

7. Read **TOTAL PIPE** and the per-leg cuts.

### Worked example

2" schedule 40 LR, three legs: 36", then 24" at 90°, then 30" at 90°, 3/32" gap:

| FIGURE            | VALUE         |
|-------------------|---------------|
| Centre to centre  | 90.00"        |
| <b>Total pipe</b> | <b>77.63"</b> |
| Elbows            | 2             |

| LEG | C2C    | CUT    |
|-----|--------|--------|
| 1   | 36.00" | 32.91" |
| 2   | 24.00" | 17.81" |
| 3   | 30.00" | 26.91" |

Leg 1 loses one takeout (it has a fitting on one end only). Leg 2 sits between two elbows and loses two. That is the whole difference, and you can see it in the numbers.

### Why the drawing is drawn the way it is

It is a **true isometric** projection — yaw  $-45^\circ$ , pitch  $35.264^\circ$ . That is the only orientation where all three axes are foreshortened equally and sit  $120^\circ$  apart, and it is what makes iso paper read as solid instead of flat.

Rotation is limited to about **260°** on purpose. The angles that are missing are the ones where the spool goes edge-on and legs hide behind each other. They are not skipped past — they are not in the range at all, so no matter how far you drag, every leg stays visible.

## Calculator

A trade calculator that understands feet, inches and fractions — and **stays in inches**.

### How to enter a dimension

Type the number, **then** the unit. Not the other way round.

| YOU WANT      | PRESS               |
|---------------|---------------------|
| 30 inches     | 3 0 Inch            |
| 6 feet        | 6 Feet              |
| 30-1/2 inches | 3 0 Inch 1 / 2 Inch |
| 6 foot 3      | 6 Feet 3 Inch       |

It stays in inches

30 Inch × 6 = gives 180", not 15 feet. The **IN** marker top left tells you which unit you are reading.

- **Feet** converts what is on display to feet and inches — 2' 6". The marker changes to **FT-IN**.
- **Inch** takes it back.
- The choice sticks until you change it, and survives a **Clear**.

The trade keys

The dark keys along the top are the pipefitting ones. **Conv** turns on the second function printed above each key.

| KEY                   | WHAT IT DOES                                |
|-----------------------|---------------------------------------------|
| Offset, Run, Travel   | Enter any two, the third solves             |
| Angle/Slope           | Angle or slope of a run                     |
| Pipe Size, Pipe Mat'l | Set the pipe for the trade keys             |
| Circle, x², √x        | Area and diameter work                      |
| Store / Rcl           | Hold a figure while you work something else |

Where people go wrong

- **Pressing the unit first.** It does nothing. Number, then unit.
- **Reading the fraction instead of the decimal.** The fraction is rounded to your setting. The decimal is exact.

Handbook

Every table from the printed book, searchable, **with its page number on it.**

Use it when you want to see the figure the calculation used rather than take it on trust. Search by what you are looking for — "reducer", "flange", "copper", "support" — and the table comes up with its source page so you can check it against the book in your truck.

## Settings

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| SETTING        | WHAT IT DOES                                     |
|----------------|--------------------------------------------------|
| Theme          | Light, dark or follow the phone                  |
| Units          | Imperial or metric                               |
| Fractions      | Off, 1/8, 1/16, 1/32, 1/64                       |
| Length readout | Inches, or feet + inches, for the calculator     |
| Default size   | Pipe every screen opens on                       |
| Weld gap       | Your standard root gap                           |
| Stock length   | Length you buy, for counting legs out of a stick |
| Updates        | What is running, and a manual check              |

Settings are stored on the phone only. Nothing leaves it.

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## Part 4 — Knowing when to trust it

### What is exact and what is not

| FIGURE                        | VALUE                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Exact                         | Every length, angle, takeout, arc and cut. These are geometry and published tables.                            |
| Estimate, and labelled as one | Fitting weights. They are geometric approximations, not vendor catalogue figures. Pipe weight itself is exact. |

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### Fractions versus decimals

The fraction is **rounded to the nearest tick** of the denominator you picked. The decimal above it is the exact figure.

On a 10-foot run at 1/16" rounding you can be out by up to 1/32". Usually nothing. On a tight flange-to-flange fit-up it is not nothing. **Cut to the decimal when the tolerance is tight.**

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### When the app refuses to answer

It refuses rather than guessing, and it tells you why:

| MESSAGE                                               | WHAT IT MEANS                                                                                          |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| "Fitting takeoffs exceed travel"                      | The fittings are longer than the space. No pipe between them. Use a smaller angle or a shorter radius. |
| "Obstruction is too close to the conduit end"         | The first bend would fall off the end. Move the mark or use a steeper angle.                           |
| "Fitting angle past 90° turns the run back on itself" | Past 90° the pipe doubles back. That is not an offset.                                                 |
| "Cut angle exceeds 22.5°"                             | Code check, ASME B31.3. Not a refusal — add a segment or get it verified.                              |
| "Thread remaining" going negative                     | Over-threaded. The pipe bottoms out before it seals.                                                   |

**A refusal is information.** It is the app telling you the job as measured cannot be built that way — which is much better news before you cut than after.

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### The habit worth keeping

For anything that matters, do this once:

1. Read **PIPE CUT**.
2. Check **Travel** against the multiplier table in Part 1.

3. Check the **takeout** looks right for the fitting in your hand.

Three seconds, and you will never wonder whether the number is good.

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*PipeFit Pro. Built for the job, tested against the book.*